Experiment 2

AIM: Setup Git and explore Commands related to Version Control System (VCS)

Step 1: Download from https://git-scm.com

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Step 2: Configuration
git config --global user.name "vatsalkakkad125750-source"
git config --global user.email "vatsal.kakkad125750@marwdiuniversity"

Step 3:
C:\Users\Vatsal>git --version
git version 2.43.0.windows.1

C:\Users\Vatsal>git init
Reinitialized existing Git repository in C:/Users/Vatsal/.git/ C:\Users\Vatsal>mkdir ost
C:\Users\Vatsal>mkdir ost
A subdirectory or file ost already exists.

C:\Users\Vatsal>git init
Reinitialized existing Git repository in C:/Users/Vatsal/.git/ C:\Users\Vatsal>cd ost

C:\Users\Vatsal\ost>git init
Initialized empty Git repository in C:/Users/Vatsal/ost/.git/

C:\Users\Vatsal\ost>git status
On branch master No commits yet
nothing to commit (create/copy files and use "git add" to track)

C:\Users\Vatsal\ost>echo sample.txt
sample.txt

C:\Users\Vatsal\ost>echo demo.py
demo.py

C:\Users\Vatsal\ost>git status
On branch master No commits yet
nothing to commit (create/copy files and use "git add" to track)


C:\Users\Vatsal\ost>dir Volume in drive C is OS
Volume Serial Number is E2F6-8AB0
Directory of C:\Users\Vatsal\ost 13-Fed-26 09:30 AM    <DIR> .
13-Fed-26 08:22 PM <DIR> ..
0 File(s)    0 bytes
2 Dir(s) 17,472,352,256 bytes free

C:\Users\Vatsal\ost>echo sample.txt
sample.txt


C:\Users\Vatsal\ost>dir Volume in drive C is OS
Volume Serial Number is E2F6-8AB0
Directory of C:\Users\Vatsal\ost 13-Fed-26 09:30 AM    <DIR> .
```

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13-Fed-26 09:30 AM <DIR> ..
0 File(s)   0 bytes
2 Dir(s) 17,472,425,984 bytes free

C:\Users\Vatsal\ost>echo Hello> sample.txt

C:\Users\Vatsal\ost>dir
Volume in drive C is OS
Volume Serial Number is E2F6-8AB0
Directory of C:\Users\Vatsal\ost 13-Fed-26 09:38 AM    <DIR> .
13-Fed-26 09:38 AM <DIR> ..
13-Fed-26 09:38 AM 7 sample.txt

     1 File(s)    7 bytes
     2 Dir(s) 17,471,066,112 bytes free

C:\Users\Vatsal\ost>type sample.txt
Hello

C:\Users\Vatsal\ost>git commit -m "Initial Commit"
On branch master Initial commit
Untracked files:
(use "git add <file>..." to include in what will be committed) sample.txt

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\Vatsal\ost>git config --global init.defaultBranch main C:\Users\Vatsal\ost>git commit -m
"Initial Commit"
On branch master Initial commit
Untracked files:
(use "git add <file>..." to include in what will be committed) sample.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Vatsal\ost>git add sample.txt

C:\Users\Vatsal\ost>git commit -m "Initial Commit"
[master (root-commit) df439d9] Initial Commit 1 file changed, 1 insertion(+)
create mode 100644 sample.txt

C:\Users\Vatsal\ost>git branch
*master

C:\Users\Vatsal\ost>git branch -m master main

C:\Users\Vatsal\ost>git branch

*main

C:\Users\Vatsal\ost>git checkout main
Already on 'main'

C:\Users\Vatsal\ost>git checkout -b dev
Switched to a new branch 'dev'

C:\Users\Vatsal\ost>echo Hello from dev>> sample.txt

C:\Users\Vatsal\ost>git add sample.txt
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C:\Users\Vatsal\ost>git commit -m "Update in dev"
[dev de8fe00] Update in dev 1 file changed, 1 insertion(+)

C:\Users\Vatsal\ost>git checkout main Switched to branch 'main'

C:\Users\Vatsal\ost>git merge dev
Updating df439d9..de8fe00 Fast-forward
sample.txt | 1 +
1 file changed, 1 insertion(+)

C:\Users\Vatsal\ost>dir Volume in drive C is OS
Volume Serial Number is E2F6-8AB0

Directory of C:\Users\Vatsal\ost 13-Fed-2609:48 AM     <DIR>.
13-Fed-26 09:44 AM <DIR>..
13-Fed-26 09:48 AM 23 sample.txt
      1 File(s)   23 bytes
      2 Dir(s) 17,468,366,848 bytes free

C:\Users\Vatsal\ost>type sample.txt
Hello
Hello from dev

C:\Users\Vatsal\ost>git remote -v

C:\Users\Vatsal\ost>git remote add origin https://github.com/vatsalkakkad125750-source/OST.git

C:\Users\Vatsal\ost>git remote -v
origin https://github.com/vatsalkakkad125750-source/OST.git (fetch) origin
https://github.com/vatsalkakkad125750-source/OST.git (push)

C:\Users\Vatsal\ost>git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/vatsalkakkad125750-source/OST.git/'

C:\Users\Vatsal\ost>git push -u origin main
info: please complete authentication in your browser... Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 452 bytes | 226.00 KiB/s, done. Total 6 (delta 0), reused 0 (delta
0), pack-reused 0
To https://github.com/vatsalkakkad125750-source/OST.git
*     [new branch] main -> main
branch 'main' set up to track 'origin/main'.
```
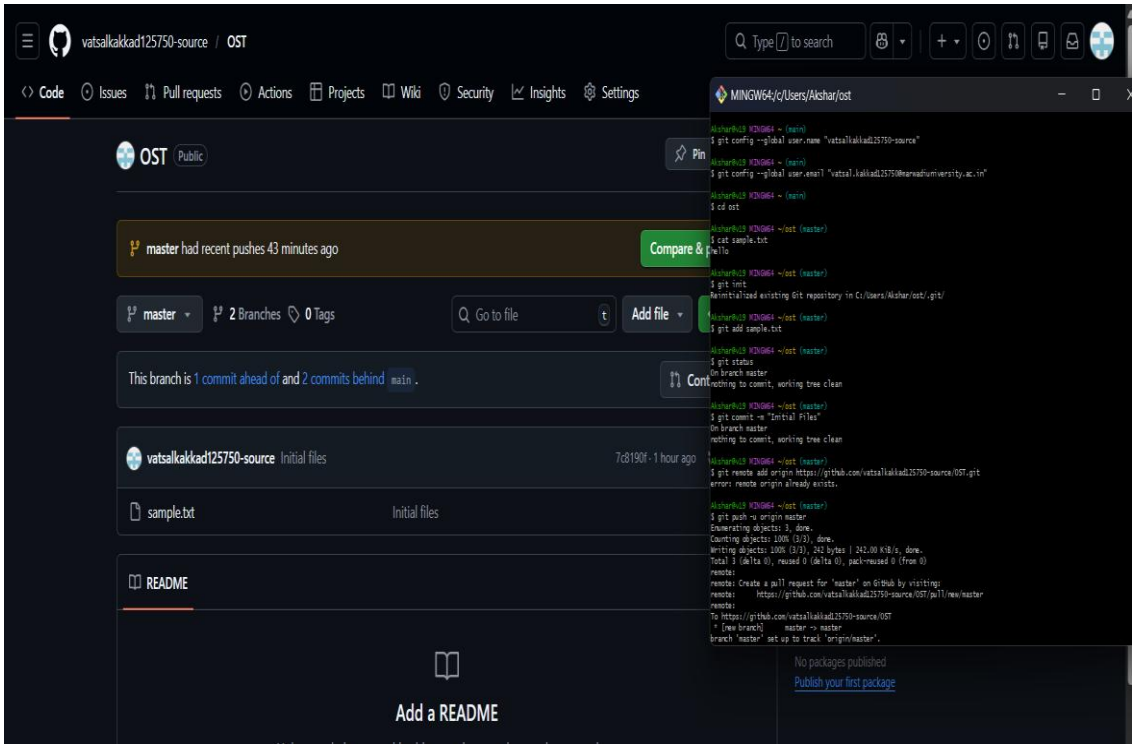

Output:


vatsalkakkad125750-source / OST
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
OST
Public
master had recent pushes 43 minutes ago
Compare &
master
2 Branches
0 Tags
Go to file
Add file
This branch is 1 commit ahead of and 2 commits behind main.
vatsalkakkad125750-source Initial files
7c8190f · 1 hour ago
sample.txt
Initial files
README
Add a README
MINGW64:/c/Users/Akshar/ost
No packages published
Publish your first package